

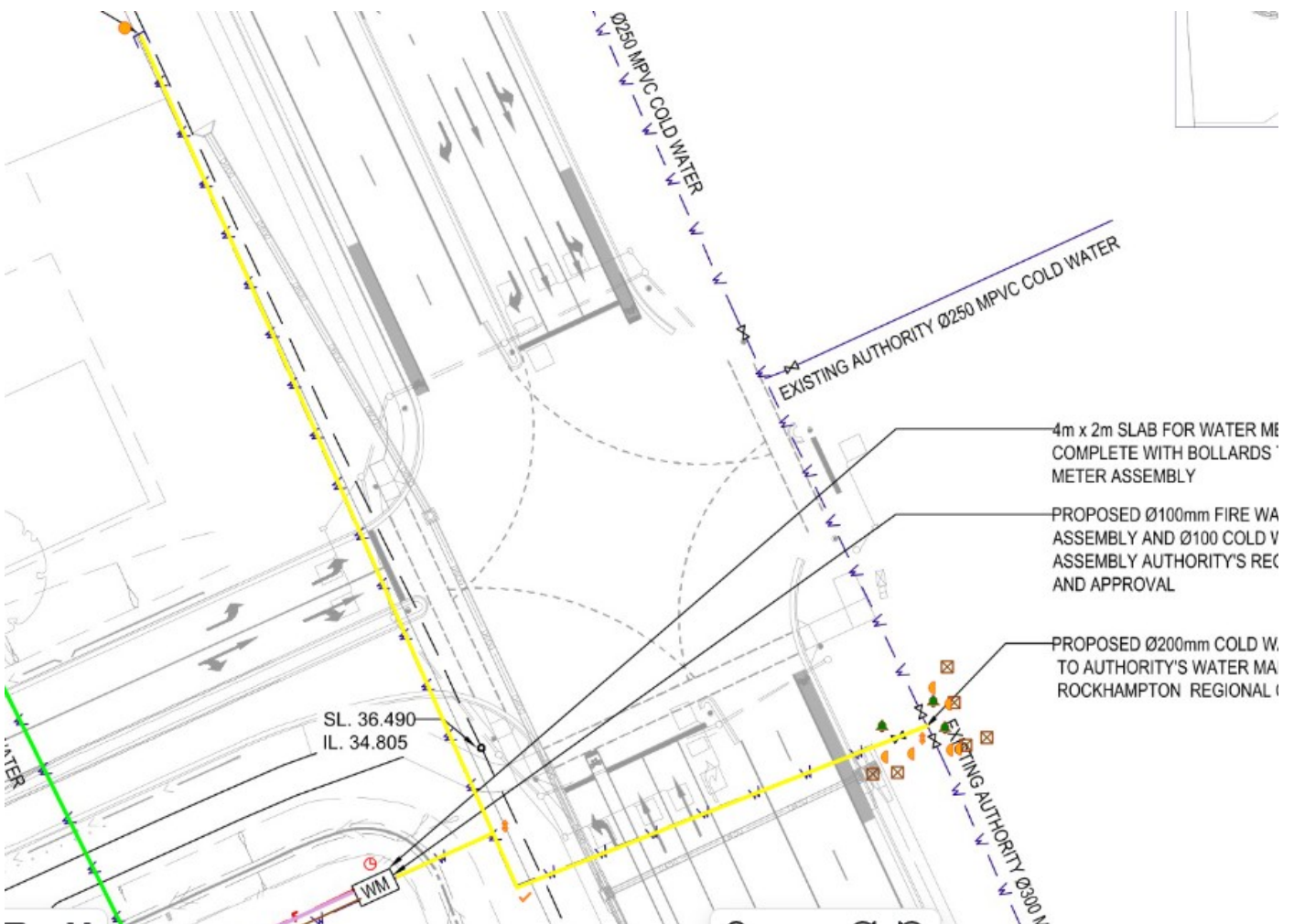
Mitch - TPS

From: Denis Zhong - TCS
Sent: Friday, 24 July 2026 5:29 PM
To: Mitch - TCS
Subject: 260724_ ROK SPORTS Hydraulic Summaries
Attachments: 1066143_ROCKHAMPTON_SPORTS_PRECINCT.pdf

-1. Site work section hydraulic , had allow for 200mm copper pipe . tee, bend and \$800 material cost for 200mm copper connection point (all the valves and fitting captured as well). The scope is from authority water main to water meter , measure in yellow.

200mm Pipe Kembla - Copper Type B	m	135
200mm copper tee	ea	2
200mm copper 90 deg bend	ea	1
200mm copper connection point	ea	1

drawing 25015-H003





10. Cold water

10.1 Generally

Supply and install all cold water pipes from the **existing Authorities water main** to all fixtures, fittings and taps requiring cold water. Include for all pipework, bends, offsets, brackets, taps and sundry equipment required for the installation.

Pipe materials shall be as follows:

Location	Material
In-ground from the Authority water main to the Boundary Water Meter	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
In-ground from Boundary Water Meter	Kembla - Copper Type B Note: When Installed in ground copper pipework to be sleeved. Refer to Specification Section: Cold & Hot Water Fixtures & Fittings
In-ground from Boundary Water Meter	Cromford – Blue Stripe PE GRADE:PN16
Cold Water Main Runs through building	Kembla - Copper Type B
Cold Water Main Runs through building	Viega – Stainless Steel - Sanpress Inox
Cold Water Rough-In (in wall)	Kembla Copper Type B
Cold Water Rough-In (in wall)	Rehau Rautitan Platinum

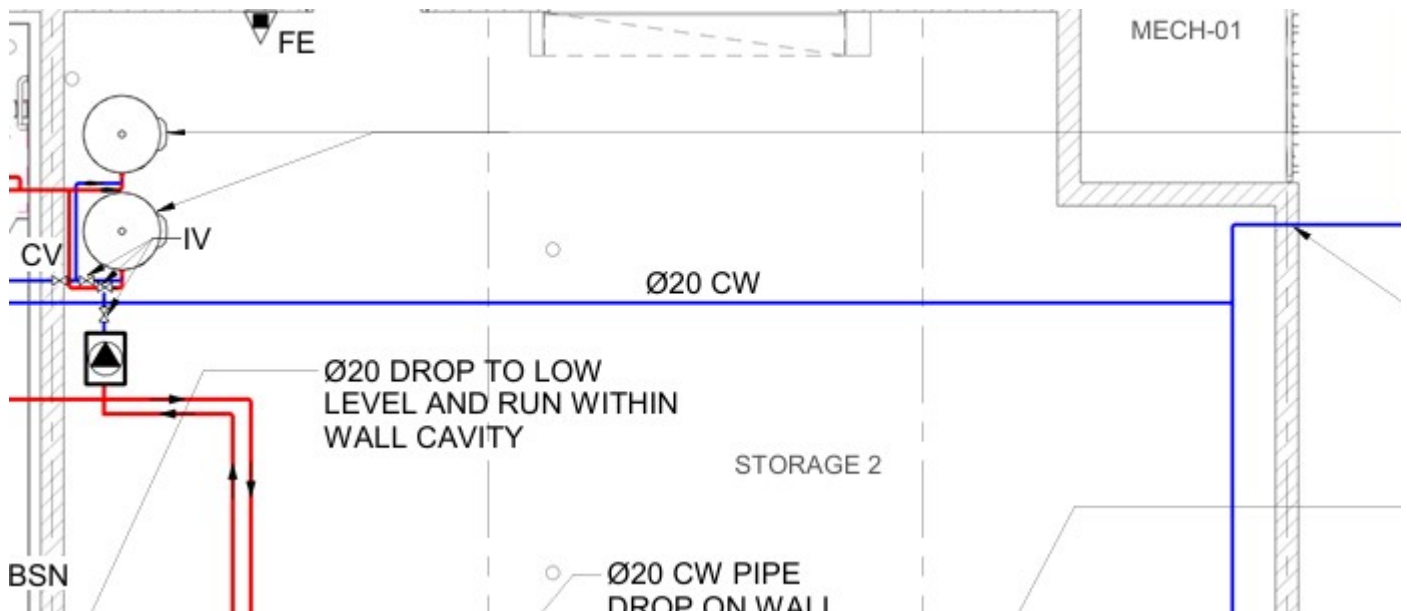
Note: All pipe sizing has been based on copper. If Rehau or Cromford is to be used, then a rating adjustment must be made in accordance with the manufacturer's data table below.

-2. Club house (NBC) / public amenities , drawing 25015-H-NBC-002, indicate above ground, **less than dn22, use pex, larger than dn22 using 316 stainless steel with press fit couplings**, I have used pex for less than dn22, and copper for larger than dn22 as talked before , pls do some allowance for the stainless steel , requested from tlink , but they did not provide rate of steel pipe .

WATER SERVICES NOTES

- WS1 REMOVE EXISTING REDUNDANT SERVICE PIPEWORK AND REDIRECT EXISTING BURIED WATER SERVICE TO NEW SERVICES GENERALLY AS SHOWN.
- WS2 ABOVE GROUND, INTERNAL DRINKING HOT & COLD WATER SYSTEM $\leq$ [SHALL BE PE-X PN16 & ARE NOMINATED ON DRAWINGS AS OUTSIDE DIA] INSTALL PE-X PLUMBING SYSTEM IN ACCORDANCE WITH THE PE-X PLUM TECHNICAL & INSTALLATION MANUAL. PIPES $>$ DN22 TO BE 316 STAINLESS STEEL WITH PRESS FIT COUPLINGS.
- WS3 ALL EXTERNAL EXPOSED COLD WATER SERVICE CONNECTION PIPES ABOVE GROUND ARE TO BE GRADE 316 STAINLESS STEEL CONNECTORS UNIONS.
- WS4 ALL EXTERNAL EXPOSED HOT WATER SERVICE CONNECTION PIPES ABOVE GROUND ARE TO BE GRADE 316 STAINLESS STEEL CONNECTORS AND UNIONS.
- WS5 INSULATION: ALL HOT WATER PIPES SHALL BE INSTALLED IN ACCORDANCE WITH THE NATIONAL CONSTRUCTION CODE OF AUSTRALIA
- WS6 ALL EXTERNAL HOSE TAPS ARE TO BE INSTALLED WITH GE ZB007 VACUUM BREAKERS IN ACCORDANCE WITH AS 3500.1.
- WS7 LOCATE HOSE TAPS AS SHOWN ON THE DRAWING.
- WS8 FOR SLAB ON GROUND AREAS, ALL WATER SERVICES TO BE RETICULATED THROUGH CEILING SPACE OR UNDER ROOF AND DROPPED FROM ABOVE. WATER SERVICES MAY BE RUN UNDER SUSPENDED FLOORS WITH RISE AND FALL WALLS. NO LONGITUDINAL SERVICES THROUGH WALLS WILL BE ACCEPTED.
- WS9 ALL INTERNAL HOT & COLD WATER RISERS & DROPPERS (INSIDE BLOCK) TO BE INSTALLED IN APPROPRIATELY SIZED CONDUIT WITH LONG RADIUS BENDS OR DUCTS FOR EASE OF REPLACEMENT.
- WS10 INSTALL HARSMITH BUILDING PRODUCTS 'FLUID APRON' (OR EQUAL) TO WALL MOUNTED TAPS & WALL ELBOWS.

-3. Club house (NBC) , drawing 25015-H-NBC-200 , assume there are 2 of RHEEM HWU and 1 of hot water pump (no spec on hot water pump) in below drawings , allow Reece rate on HWU and Hot water pump (reece quote the pump with higher duty) .



-4. Need to allow for the hanger rods for the pipe work as below , the soil is high active according to geo report .

Site work section hydraulic , drawing 25015-H0002

9. SEISMIC RESTRAINT OF NON-STRUCTURAL COMPONENTS

ALL PIPEWORK SHALL BE PROVIDED WITH SEISMIC RESTRAINTS COMPLYING WITH AND LONGITUDINALLY STABILISATION. HANGER RODS SHALL ALSO BE PROVIDED WITH BUCKLING DURING SEISMIC EVENTS.

DUCTILE MATERIAL SPACING

THE MAXIMUM ALLOWABLE BRACE SPACING FOR PIPING CONSTRUCTED OF DUCTILE ALUMINIUM) ARE:

- 12M FOR TRANSVERSE BRACING (PIPING LARGER THAN 65 MM DIA.);
- 24M FOR LONGITUDINAL BRACING (PIPING LARGER THAN 65 MM DIA.);
- 9M FOR TRANSVERSE BRACING (PIPING SMALLER THAN 65 MM DIA.);
- 18M FOR LONGITUDINAL BRACING (PIPING SMALLER THAN 65 MM DIA.).

NON-DUCTILE MATERIAL SPACING

THE MAXIMUM ALLOWABLE BRACE SPACING LIMITS FOR PIPING NON-DUCTILE MATERIAL

- 6 M FOR TRANSVERSE BRACING;
- 12 M FOR LONGITUDINAL BRACING.

10. CONCRETE

In project doc VOL 38 hydraulic service spec

temporarily sealed off with patented plastic caps.

3.14 Fixing & supporting of pipes

3.14.1 General

Where required, as per the seismic restraint section of the specification, provided with seismic restraints complying with AS1170.4, providing transversely and longitudinally. Hanger rods shall also be provided to prevent buckling during seismic events. Supports and fixings to AS1170 by suitably qualified persons and be suitable for the application and material.

Pipework and fittings shall be:

- a. Only fixed in approved locations.
- b. Adequately secured to the structure to support the pipework under load with a safety margin of 2:1.
- c. Kept clear of structure and other services.
- d. Provided with 0.75mm galvanised sheet metal sleeves where passing through structure. Pack space between pipe and sleeve with fire rated material of structure through which pipe passes.
- e. Installed to allow for adequate expansion and contraction without stressing pipes or joints.
- f. Fixed on hanger brackets to allow adjustment for fall.
- g. Not be fixed with explosive power tools.
- h. Cleaned of all cement dropping on completion.

3.14.2 Materials

- a. Use purpose-made galvanised mild steel channel equal to complete with purpose-made fittings for multiple parallel pipes. End caps equal to "Unistrut A1280" on exposed channels.



Figure 1 P1000T Channel

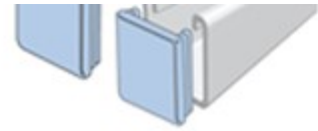


Figure 2 A1280 End Bracket

- b. Use two-piece brackets equal to “Unistrut Pipe Clamps” to secure and rubber cushioning equal to “Unistrut Unicushion” between brackets where Insulation/Lagging is not required.

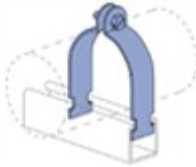


Figure 3 Two Piece Pipe Clamps

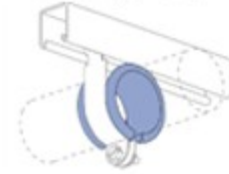


Figure 4 Unicushion

- c. Use Timber Blocks between pipework and bracketing when installed.



Figure 5 Timber Block



Figure 6 Typical Timber Block Installation

- d. Flexistrut

- i. Use brackets equal to “Flexistrut Nut Clamp” for service pipes where Insulation/Lagging is installed.
- ii. Use clips equal to “Flexistrut Acoustic Pipe Clips” for pressure service pipes where Insulation/Lagging is not required.



Figure 7 Nut Clamp



Figure 8 Acoustic Pipe Clip

- e. Use slide guide equal to “Flexistrut Heavy Duty Slide Guide” for hot/warm water flow and return pipework.





Regards,

Denis Zhong

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